MIL-H-21303D
8 FEBRUARY 1973
SUPERSEDING
MIL-H-21303C
27 January 1971

MILITARY SPECIFICATION

HOT CUPS, LIQUID, ELECTRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of electrically heated food-warming cups.

1.2 Classification. Hot cups shall be of the following types, as specified (see 6.2):

Type I - Nominal, 28 Volt, DC
Type II - Nominal, 115 Volt, AC, 400 Hz

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-N-281	Nickel-Copper Alloy Bar, Plate, Rod, Sheet, Wire, Forgings and Structural and Special Shaped Sections
QQ-C-576	Copper Flat Products, with Slit, Slit and Edge-Rolled, Sheared, Sawed, or Machined Edges (Plate, Bar, Sheet and Strip)
QQ-S-698	Steel, Sheet and Strip, Low Carbon
QQ-S-766	Steel Plates, Sheet, and Strip-Corrosion Resisting

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000	Drawings, Engineering, and Associated Lists
MIL-B-7525	Bracket and Receptacle, Hot Cup, Airborne

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

FSC 7310

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MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-454	Standard General Requirements for Electrical Equipment
MIL-STD-794	Parts and Equipment, Procedure for Packing and Packaging of
MIL-STD-882	System Safety Program, for Systems and Associated Subsystems and Equipment: Requirements for

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL SANITATION FOUNDATION (NSF)

Standard No 4 Commercial Cooking and Warming Equipment List of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, P. O. Box 1468, Ann Arbor, MI 48106.)

UNDERWRITERS' LABORATORIES, INC.

Standard No 197 Commercial Cooking Equipment

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 207 East Ohio St., Chicago, IL 60611; 1655 Scott Blvd., Santa Clara, CA 95059; 2500 Dundee Rd., Box 247, Northbrook, IL 60062.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

ICS 1-109 Tests

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 155 East 44th St., New York, NY 10017.)

DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION

Occupational Safety and Health Act (OSHA) Standards, with Supplemental Revisions

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), before production is commenced, a sample of the required hot cup covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for examination and approval in accordance with 4.2. Approval of the preproduction sample authorizes the commencement of

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production, but does not relieve the supplier of the responsibility for compliance with the detail provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

* 3.2 Standards.

* 3.2.1 UL. The hot cup shall conform to the applicable requirements of UL Standard No 197.

* 3.2.1.1 Certification. Prior to the approval of the preproduction sample, or prior to commencing production, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the hot cup he proposes to furnish under this specification meets the applicable requirements of UL Standard No 197. Acceptable evidence of meeting the applicable requirements of these standards shall be the UL label, or the UL listing mark, or a certified test report from a recognized, independent testing laboratory acceptable to the Government, indicating that the hot cup has been tested and conforms to the applicable requirements of UL Standard No 197.

* 3.2.2 NSF. The hot cup shall conform to the applicable requirements of NSF Standard No.4.

* 3.2.2.1 Certification. The supplier shall submit satisfactory evidence to the contracting officer or his representative that the hot cup he proposes to furnish under this specification meets the applicable requirements of NSF Standard No 4 as follows:

(1) A listing in the current edition of NSF "Listing of Food Service Equipment" or display of the NSF Seal on the finished hot cup shall be submitted prior to approval of the preproduction sample, or

(2) A certified test report from a recognized independent testing facility acceptable to the Government indicating that the hot cup has been tested and conforms to the applicable sections of NSF Standard No 4 shall be submitted prior to approval of the preproduction sample.

3.3 Materials. Materials shall conform to the applicable specifications and requirements specified herein. Selection of materials not specifically defined shall be of the quality normally used in the manufacturer's standard commercial item.

3.3.1 Steel, carbon sheet. Carbon steel sheet shall conform to cold-rolled commercial quality of QQ-S-698.

3.3.2 Steel, corrosion resisting. Corrosion resisting steel shall be in accordance with classes 302, 303, 304, or 305, condition A of QQ-S-766.

3.3.3 Copper. Copper sheet shall conform to QQ-C-576.

3.3.4 Plastic. Plastics shall be phenol formaldehyde or polycarbonate types or the manufacturer's standard type, provided it meets the requirements specified herein. The color shall be black.

3.3.5 Nickel. Nickel sheet shall conform to QQ-N-281, Class A.

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3.4 Design. Hot cups specified herein shall consist of two basic designs, Type I, designed to operate on 28 volt DC, and Type II, designed to operate on 115 volt AC, 400 Hz. Both Type I and II hot cups shall be suitable for use in appropriate hot cup brackets specified in MIL-B-7525. The hot cup designs shall include a thermostat to control the following functions:

a. Provide protection when operated in a dry condition.

b. Alleviate the necessity of attention by crewmen during flight operations.

c. Control "boil-over" when used in an aircraft cabin with pressurization corresponding to 5,000 feet elevation above sea level. The brim capacity of both Type I and II hot cups shall be 35 ±2 fluid ounces.

3.5 Construction. Details not specifically defined hereinafter shall be in accordance with the manufacturer's standard practice. Type I and II hot cups shall generally conform to Figures 1 and 2, respectively. Each type hot cup shall consist of the following: An outer shell, an inner shell, a lid assembly, a handle, a two-pronged electrical plug, a heating element, and a thermostat.

3.5.1 Outer shell. The outer shell shall be constructed of corrosion resisting steel conforming to 3.3.2 or of carbon steel conforming to 3.3.1.

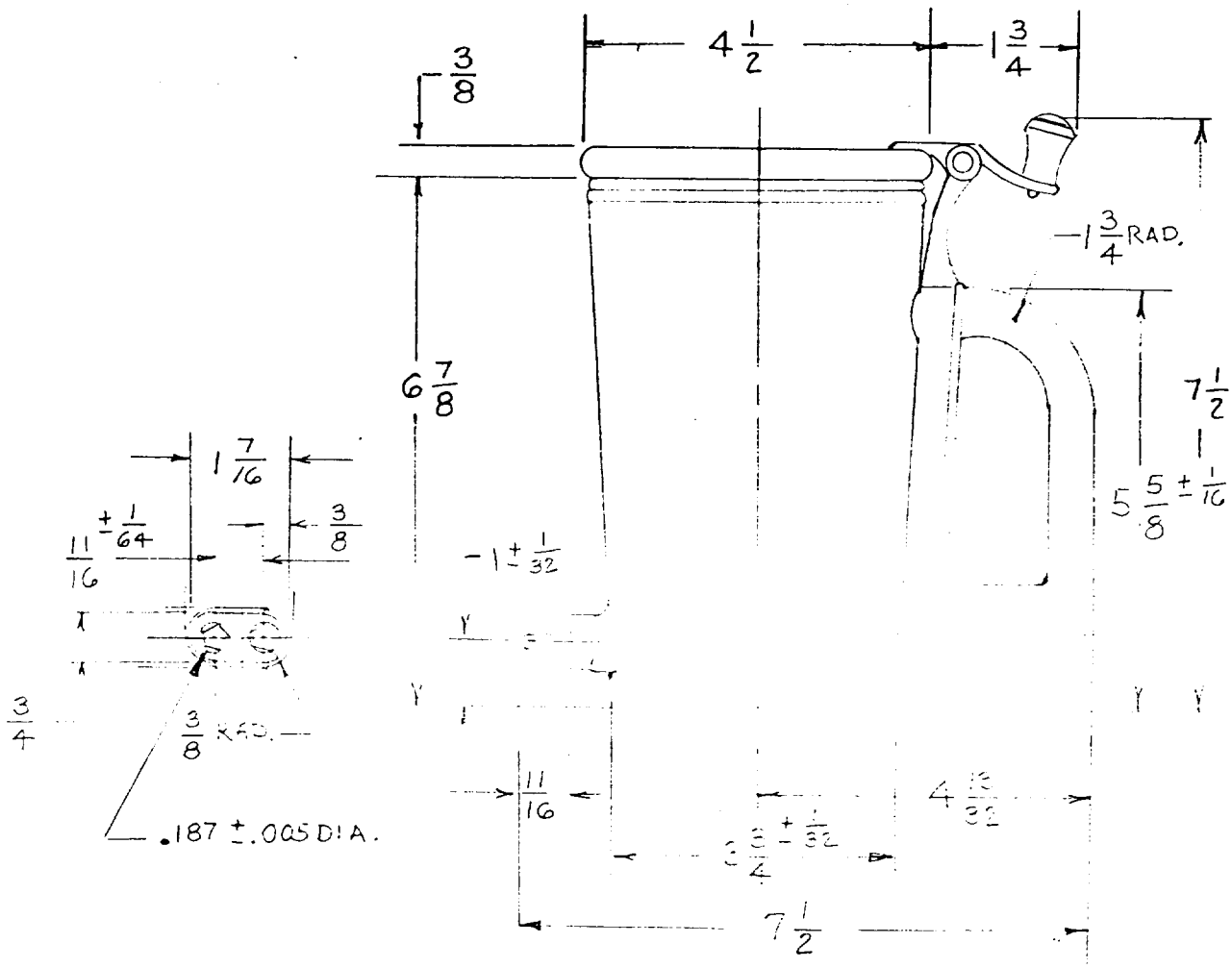
3.5.2 Inner shell. The inner shell shall be constructed of corrosion resisting steel conforming to 3.3.2, copper conforming to 3.3.3, or nickel conforming to 3.3.5 (see 6.2).

3.5.3 Lid assembly. The periphery of the lid shall be rounded as shown in Figures 1 and 2, in order to provide good closure and strength. The lid shall be shaped to return all condensate and splash to the inside of the cup. The lid shall be fastened securely to a hinge arm. The other hinge arm shall be fastened to the cup. All hinge parts shall be sufficiently strong to permit repeated slamming of the lid without breakage. The lid assembly shall be replaceable.

3.5.4 Handle. Each hot cup shall be equipped with a handle that incorporates a guard to protect the user's hand from the outer shell heat. The combination handle and guard shall be of plastic in accordance with 3.3.4 and shall be attached to the outer shell in such a manner so as not to create a safety or sanitation problem. The combination handle and guard shall be replaceable in the event of fracture.

* 3.5.5 Electrical plug. The electrical plug shall be located as indicated on Figures 1 and 2 for Type I and II hot cups respectively. Where the electrical plug connections penetrate the outer shell, they shall be potted with a suitable compound to prevent leakage of water into the area between the inner and outer shells. The potting compound shall be heat resistant and shall not be brittle or susceptible to cracking or leakage. Care must be exercised to prevent using excessive amounts of the potting compound that would prevent the plug from mating with its intended receptacle. The electrical plugs shall be protected by a guard as shown in Figures 1 and 2.

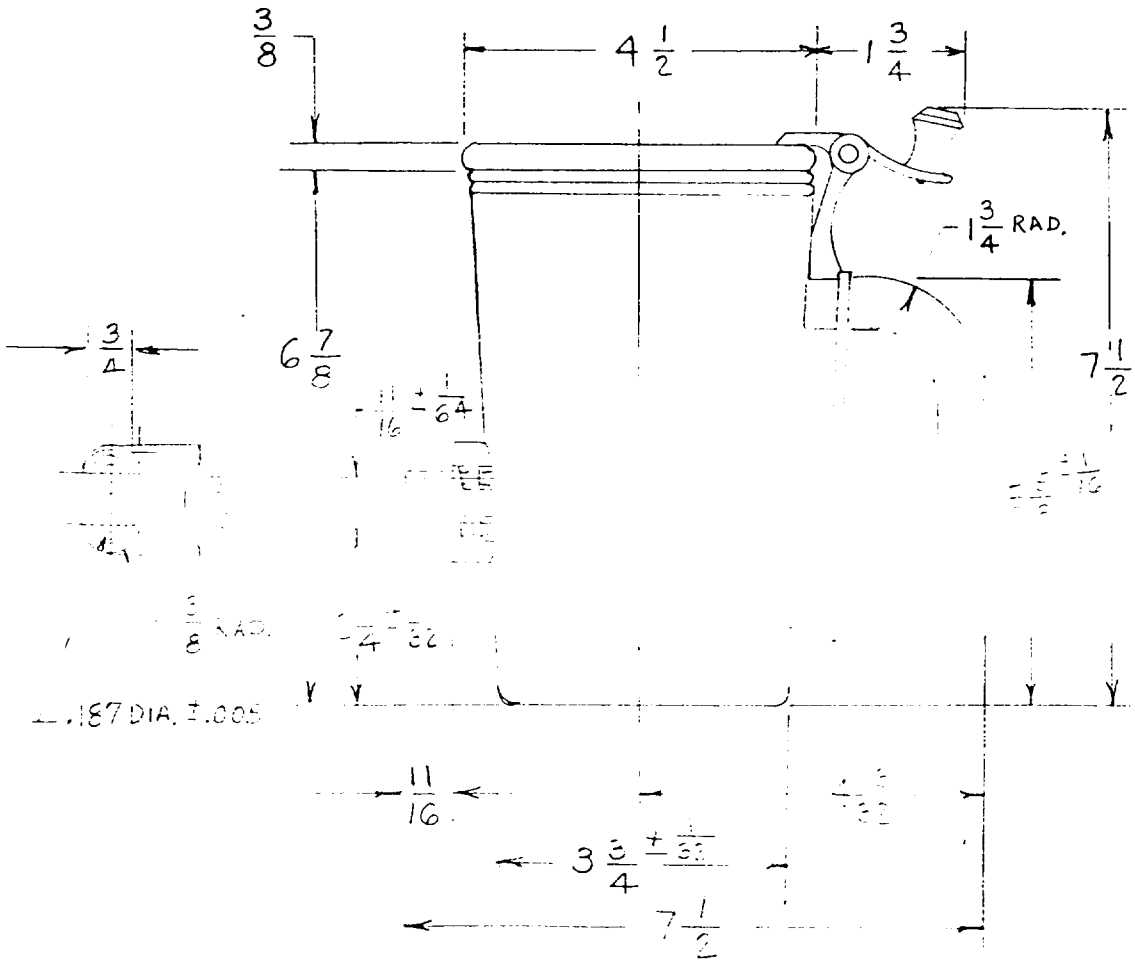
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DIMENSIONS IN INCHES. THIS FIGURE IS NOT RESTRICTIVE EXCEPT FOR THOSE DIMENSIONS SHOWN WITH TOLERANCES.

FIGURE I: DIMENSIONAL DETAILS OF THE TYPE I FOOD WARMING CUP

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DIMENSIONS IN INCHES: THIS FIGURE IS NOT RESTRICTIVE EXCEPT FOR THOSE DIMENSIONS SHOWN WITH TOLERANCES.

FIGURE II: DIMENSIONAL DETAILS OF THE TYPE II FOOD WARMING CUP

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3.5.6 Heating element. The heating elements shall be located between the inner and outer shell and shall meet the performance characteristics specified in 3.6.3.

3.5.7 Thermostat. The thermostat shall meet the performance characteristics specified in 3.6.4.

3.5.8 Cord assembly. When Type II hot cups specified herein are not to be used in hot cup brackets delineated in MIL-B-7525, a cord assembly shall be specified (see 6.2). The cord assembly shall consist of a two-connector, 16-2 HPN flexible heater cord; a male, two-prong plug suitable for connection to a standard, non-locking type receptacle; and a heater type female plug suitable for attachment to the hot cup connections shown in Figures 1 and 2. The cord length shall be specified by the procuring agency. Cords shall be used on Type II cups only.

3.5.9 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-1000 shall govern changes in the manufacturer's part number.

3.6 Performance.

3.6.1 Lid assembly. The lid shall be designed to open 80°, +10°, -0° and shall be freely hinged to drop to a closed position when released.

3.6.2 Handle. The handle shall be heat resistant such that it can be hand held during all modes of operation.

3.6.3 Heating element. The power consumption for Type I hot cup heating elements shall not exceed 600 ±30 watts, and shall operate satisfactorily on voltages from 20 to 29 volts DC. The power consumption for Type II hot cup heating elements shall not exceed 650 ±30 watts and shall operate satisfactorily on voltages from 102 to 124 volts rms with frequency limits of 380 to 420 Hz. The heating elements and the two contact pins for both type cups shall be electrically insulated from the inner and outer shell so as to withstand the dielectric strength test of 4.4.2.

3.6.4 Thermostat. During the heating cycle, the thermostat shall interrupt the current when the temperature of the liquid contained within the hot cup is between 170° F and 190° F. When the thermostat opens the thermal overshoot temperature of the liquid in the hot cup shall not exceed 205° F. The water temperature shall not drop to less than 160° F before a reheating cycle begins.

* 3.7 Plating and finish. When corrosion resisting steel or nickel alloy is used for hot cup construction, the inside surface of the inner shell, the lid assembly, and the outside of the outer shell shall have a polished finish to meet the requirements of NSF Standard No 4 for food service equipment. When carbon steel and copper are used for construction, the inside surface at the inner shell, the lid assembly, and the outside surface of the outer shell shall be plated with a high quality, commercial, bright chromium. The chromium plating shall be over nickel plating over copper and the resulting finish shall conform to NSF Standard No 4.

3.8 Identification of product. Each hot cup shall be permanently marked for identification in accordance with MIL-STD-130.

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3.9 Workmanship. The workmanship shall be in accordance with MIL-STD-454, requirement 9.

3.10 Safety. The hot cups shall be consistent with the applicable "OSHA" standards, and shall be free of Category III and Category IV hazards as determined in accordance with MIL-STD-882.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. When a preproduction sample is required, it shall be examined for the defects listed in Table II, dimensions specified in Figures 1 or 2, and tested in accordance with the tests listed in 4.4.1 through 4.4.7.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105 except where otherwise indicated herein.

4.3.1 Component and material inspection. In accordance with 4.1, the supplier is responsible for insuring that components and materials used are manufactured, tested, and inspected in accordance with all the requirements of referenced specifications, drawings and standards, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 Process inspection. Examination shall be made at appropriate stages of manufacture to determine compliance with requirements in Table I. Whenever nonconformance is noted, correction shall be made to affected items in process.

TABLE I
PROCESS EXAMINATION

OPERATION	REQUIREMENT
Handle Attachment	3.5.4
Electrical Plug Potting	3.5.5
Plating and Finish	3.7

4.3.3 End item inspection. The lot shall consist of all hot cups of one type offered for inspection at one time.

4.3.3.1 Visual inspection. Inspection shall be made of the hot cups for the defects listed in Table II. The inspection level shall be level II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE II
* CLASSIFICATION OF DEFECTS

	Examine	Defect	Classification Major	Minor
1.	Design, Construction and workmanship	Combination handle and guard not as specified.	X	
		Lid assembly not as specified.	X	
		Component missing, fractured, dented, or malformed.	X	
		Cord assembly (when applicable) damaged.	X	
		Electric pins or thumb lever bent.	X	
		Poor joint between inner and outer shell.	X	
		Poor potting or excess potting around electrical pins.	X	
		Pin guard joint not smooth.		X
2.	Materials	Not as specified.	X	
3.	Surfaces	Rough or not polished.	X	
		Plating chipped or peeling.	X	
		Scratched (small).		X
		Color of thumb lever tip and handle not black.		X
4.	Dimensions	Not as specified in Fig 1 or 2, or does not fit bracket of MIL-B-7525	X	
5.	Marking	Not as specified.	X	

4.3.3.2 End product testing. All hot cups shall be tested as specified in paragraphs 4.4.3 and 4.4.6. Using an inspection level of S-4 and an AQL of 4.0 expressed in terms of defects per hundred units, sample hot cups shall be tested in accordance with para 4.4.2, 4.4.4, 4.4.5, 4.4.7, and 4.4.8.

4.4 Tests.

4.4.1 Handle test. The handle shall be hand-held while the hot cup is operated through a complete heat-up cycle as determined by high limit cut-off of the thermostat to demonstrate that the handle and guard protect one from burning as required by 3.6.2.

*** 4.4.2 Dielectric strength. All hot cups shall be tested for dielectric strength in accordance with NEMA standards, Part ICS 1-109. Failure to pass this test shall be cause for rejection of that cup.**

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4.4.3 Potting and cup leak test. The sample hot cups at ambient conditions (80° F maximum) shall be immersed totally in water which is heated to 200° F ±5° F for a period of 10 minutes. Leakage (indicated by air bubbles) between the inner and outer shells or around the electrical pin potting shall be cause for rejection of that particular cup.

4.4.4 Net content. The sample cup shall be filled with water and the capacity shall conform to the requirements specified in 3.4.

4.4.5 Operation. All hot cups shall be filled with 30 ounces of water at 70° ±5° F and operated at rated voltage. The lid of the cup shall be closed. The temperature of the water shall reach 175° ±5° F in 12 ±2 minutes. No fluid shall escape from inside the cup during the heating of the fluid. After the maximum operating temperature has been reached, the lid shall be opened to the maximum position. None of the condensate on the inside of the lid shall drip outside the cup. The thermostat shall meet the requirements of 3.6.4.

4.4.6 Electrical characteristics. Immediately following the test specified in 4.4.4 the sample hot cups shall be tested to verify that they meet the electrical characteristics and power consumption required in paragraph 3.6.3 except that Type II hot cups may be tested using 115 V, 60 Hz in lieu of 380 to 420 Hz.

4.4.7 Dry operation. The sample cups shall be connected to a power source and permitted to run continuously, dry, for 24 hours. Failure shall be cause to reject the entire lot.

4.4.8 Preparation for delivery. The inspection shall ascertain that the preparation for delivery requirements of hot cups shall be in accordance with this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging of hot cups shall be level A, B, or C as specified (see 6.2) in accordance with MIL-STD-794, utilizing method III of MIL-P-116.

5.2 Packing. Packing of hot cups shall be level A, B, or C as specified (see 6.2) in accordance with MIL-STD-794.

5.3 Marking. Marking of all containers shall be in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The hot cups covered by this specification are intended to be used in flight, with cabin pressurization approximately equivalent to 5000 feet elevation, for heating liquids such as hot chocolate drinks, coffee, and soups. They shall be suitable for use in hot cup brackets conforming to MIL-B-7525.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type of cup required (see 1.2).
c. Preproduction sample (not required) (see 3.1).

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d. Inner cup material (see 3.5.2).
e. Cord assembly; if required, specify length (see 3.5.8).
f. Selection of applicable levels of packaging and packing (see 5).

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - GL
Navy - AS
Air Force - 84

Preparing activity:
Air Force - 84

Review activities:
Air Force - 45
Army - MD

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-H-21303D; Hot Cups, Liquid, Electric

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE. AF-WP-O-MAY 64 505

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